Inverse Takeoff Kernels:
System Identification for Recursive Redundancy

Jonathan R. Landers

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Abstract

Velocity takeoff kernels describe how redundant work becomes visible in a recursive computation before memoization collapses the recursion tree into a directed acyclic graph of distinct subproblems. This paper develops the inverse question. Given an observed redundancy takeoff curve, what can be inferred about the recursive mechanisms that produced it? The answer is deliberately two-stage. First, under a finite-exponential response model, the observed takeoff profile admits a modal expansion whose poles and amplitudes can be recovered exactly from noiseless samples by Hankel, Prony, matrix-pencil, or minimal-recurrence methods, and approximately from noisy traces by regularized system identification. A dominance theorem then explains when recovered modes become visible in the composite response: the largest-radius mode controls the tail, while pairwise crossover estimates locate the handoff between short-lived and slow-settling components. Second, those response modes constrain but do not generally identify literal branch sources. A combined recurrence may be split into fast and slow channels in many observationally equivalent ways. The resulting principle is that takeoff identifies response modes before it identifies source mechanisms. Source-level reconstruction becomes meaningful only after adding structure such as bounded lag, nonnegative integer branch counts, sparsity, separated supports, or multiple instrumented traces.

1 Introduction

The forward theory of takeoff kernels starts with a recursive schema and asks what redundancy response it generates. In a finite-lag model,

$$N_n = 1 + \sum_{j \in J} a_j N_{n-j},$$

the tree velocity

$$u_n = \log N_{n+1} - \log N_n$$

approaches a terminal speed α. The normalized profile

$$F_n = \frac{u_n}{\alpha}$$

is a step response, and the takeoff kernel

$$\kappa_n = F_n - F_{n-1}, \qquad F_{-1} = 0,$$

records the finite shape of redundancy onset. The subdominant roots of the recurrence determine whether the response is immediate, delayed, monotone, oscillatory, mixed, or slowly settling.

That forward picture suggests a sharper empirical question. If a recursive program is instrumented and its redundant work is observed, can the response be read backward? A trace of N_n is not just a cost curve. After normalization, it is a finite response shape. It may contain a fast initial component, a slow tail, a small alternating ripple, or a near-periodic settling mode. The inverse problem asks how much of that structure can be recovered from the curve itself:

$$\text{observed takeoff} \longrightarrow \text{modal decomposition} \\ \longrightarrow \text{candidate recurrence mechanisms.}$$

The analogy is system identification, but the object being identified is not a physical plant. It is the redundant response of a recursion tree before the memoized quotient removes repeated labels. The observed profile is therefore closer to a measured impulse or step response than to a conventional runtime bound.

The main caution is identifiability. A composite recurrence may have two channels,

$$a_j = b_j + c_j,$$

where b_j is a fast channel and c_j is a slow channel. The observed tree sequence depends on the combined coefficients a_j, not on the chosen split. Thus a takeoff trace can identify response modes more directly than it can identify literal source branches. This is the paper's organizing distinction: modal structure is the primary observable; source structure is a constrained interpretation layered on top of it.

Proposed contributions. The paper develops six pieces.

1. Define inverse takeoff as a recovery problem for observed redundancy velocity profiles.
2. Prove an exact modal recovery theorem in the noiseless finite-exponential case.
3. Characterize when recovered modes become apparent in the composite response through tail dominance and crossover estimates.
4. Prove a nonidentifiability result for latent branch-channel decompositions.
5. Develop constrained reconstruction criteria for plausible recursive mechanisms.
6. Demonstrate the method on synthetic and real recursive traces.

Planned Figure 1: inverse takeoff pipeline. Show the measured curve N_n, the normalized residual $Y_n = F_n - 1$, the Hankel singular spectrum used to choose modal order, the recovered modes θ_r, and the final constrained recurrence candidates. The point of the figure is to separate response identification from source interpretation.

Figure 1: Conceptual pipeline for inverse takeoff.

2 Observed Takeoff Data

Let N_n be the number of calls in the naive recursion tree and M_n the number of distinct subproblem labels under memoization. Define

$$u_n = \log N_{n+1} - \log N_n, \qquad w_n = \log M_{n+1} - \log M_n,$$

and

$$v_n = u_n - w_n.$$

The sequence u_n is the tree-side velocity; it records how quickly recursive multiplicity is being created. The sequence w_n is the state-space velocity; it records how quickly the memoized DAG itself is growing. The overlap velocity v_n is their difference. This distinction is important in the inverse setting because two different response curves can produce a visually simple difference.

When $u_n \to \alpha > 0$, the tree takeoff profile is

$$F_n = \frac{u_n}{\alpha}.$$

When $w_n \to \beta > 0$, the DAG profile is

$$G_n = \frac{w_n}{\beta}.$$

The overlap velocity obeys

$$v_n - (\alpha - \beta) = \alpha(F_n - 1) - \beta(G_n - 1).$$

Thus, if only v_n is observed, tree and DAG modes may cancel or mask one another. The inverse problem is best posed using N_n and M_n separately when those traces are available.

The cleanest data product is therefore

$$\mathcal{D}_T = \{(N_n, M_n) : 0 \leq n \leq T\}.$$

From $\mathcal{D}_T$, one computes u_n, w_n, v_n, estimates terminal speeds α, β, and then forms normalized residuals. If only the tree trace is available, the main residual is

$$Y_n = F_n - 1.$$

If only overlap velocity is available, the measured residual is instead

$$Z_n = v_n - (\widehat{\alpha} - \widehat{\beta}),$$

which should not be interpreted as a pure tree mode unless w_n is known to settle faster or be negligible.

In measured data, write

$$\widehat{Y}_n = Y_n + \varepsilon_n, \qquad n = n_0, \ldots, n_1,$$

Planned Figure 2: data normalization. Panel (a): raw N_n on a log scale. Panel (b): tree velocity u_n with the fitted terminal speed $\widehat{\alpha}$. Panel (c): residual $Y_n = \widehat{F}_n - 1$ after a warmup cutoff. Panel (d): sensitivity of the residual to the chosen tail window.

Figure 2: From call counts to an inverse-takeoff residual.

where ε_n includes measurement noise, terminal-speed estimation error, boundary effects, and model misspecification. The finite prefix is often contaminated by base-case conventions, so the inverse procedure should allow a warmup cutoff n_0. The tail, meanwhile, may be too short to expose a slow mode. A credible inverse analysis must therefore report the chosen observation window, not only the fitted parameters.

Estimating terminal speed. For a tree trace, a simple estimator is a tail average

$$\widehat{\alpha} = \frac{1}{|\mathcal{T}|} \sum_{n \in \mathcal{T}} u_n$$

over a stable tail window $\mathcal{T}$. A more robust version fits

$$\log N_n = c + \alpha n + \delta_n$$

on the tail and then studies the residual velocity. The inverse method should be run under several plausible tail windows; a mode that disappears under small window shifts is better treated as a numerical artifact than as a structural claim.

3 Three Inverse Problems

Problem 1 (Modal inverse takeoff)**.** *Given samples $Y_0, \ldots, Y_T$, recover a small modal model*

$$F_n - 1 \approx \sum_{r=1}^{m} c_r \theta_r^n.$$

For complex conjugate modes this can be written as a sum of damped sinusoids,

$$A_r r_r^n \cos(\omega_r n + \phi_r).$$

The target is not merely interpolation. The fitted modes should predict unobserved tail values, explain the observed settling envelope, and remain stable under modest changes of the observation window.

Problem 2 (Combined recurrence reconstruction)**.** *Given $N_0, \ldots, N_T$, recover a bounded-lag recurrence*

$$N_n \approx 1 + \sum_{j=1}^{L} a_j N_{n-j}, \qquad a_j \geq 0,$$

preferably with $a_j \in \mathbb{Z}_{\geq 0}$ when the model represents branch counts. This problem asks for a combined mechanism. It does not yet ask which branch or subroutine supplied each lag.

Problem 3 (Latent channel decomposition)**.** *Given recovered combined coefficients a_j, recover a decomposition*

$$a_j = b_j + c_j$$

into fast, slow, or otherwise interpretable channels.

These problems are ordered by identifiability. Modal inverse takeoff is the best-posed. Combined recurrence reconstruction is plausible under bounded-lag and positivity assumptions. Latent channel decomposition is not identifiable in general and requires additional structure.

The ordering also suggests how results should be reported. The first output is a modal table

$$(\theta_r, c_r) = (r_r e^{i\omega_r}, c_r), \qquad r = 1, \dots, m,$$

with damping r_r, frequency ω_r, phase, and uncertainty. The second output is a small set of candidate recurrences a, each with a forward-simulated takeoff curve. The final output, if attempted, is a source story such as "one local branch and one delayed branch." Only the first of these is directly identified by a single trace.

4 Exact Modal Recovery

Suppose the takeoff tail has the finite-exponential form

$$y_n := F_n - 1 = \sum_{r=1}^{m} c_r \theta_r^n,$$

with distinct nonzero modes θ_r and nonzero amplitudes c_r. Define the Hankel matrix

$$H_{ij}^{(s)} = y_{s+i+j}, \qquad 0 \le i, j \le q - 1.$$

Let V_q be the $q \times m$ Vandermonde matrix

$$(V_q)_{ir} = \theta_r^i, \qquad 0 \le i \le q - 1, \quad 1 \le r \le m.$$

Then

$$H^{(s)} = V_q \operatorname{diag}(c_1\theta_1^s, \dots, c_m\theta_m^s) V_q^T.$$

Consequently, for $q \ge m$, the infinite noiseless Hankel matrix has rank m. This is the algebraic signature of a finite modal response: the data live on an m-dimensional recurrence model even though the observed trace may look like a composite curve.

Theorem 1 (Noiseless modal recovery)**.** *Assume*

$$y_n = \sum_{r=1}^{m} c_r \theta_r^n$$

with distinct nonzero θ_r and nonzero c_r. Then the $2m$ consecutive samples $y_0, \dots, y_{2m-1}$ determine the monic annihilating polynomial

$$p(t) = t^m + p_{m-1}t^{m-1} + \cdots + p_0 = \prod_{r=1}^{m} (t - \theta_r),$$

uniquely. The modes θ_r are the roots of p, and the amplitudes c_r are recovered uniquely by a Vandermonde system.

Proof. The sequence satisfies

$$y_{n+m} + p_{m-1}y_{n+m-1} + \cdots + p_0 y_n = 0$$

for every n, because each θ_r is a root of p. For $n = 0, \dots, m-1$, this gives the Hankel system

$$\begin{bmatrix} y_0 & y_1 & \cdots & y_{m-1} \\ y_1 & y_2 & \cdots & y_m \\ \vdots & \vdots & \ddots & \vdots \\ y_{m-1} & y_m & \cdots & y_{2m-2} \end{bmatrix} \begin{bmatrix} p_0 \\ p_1 \\ \vdots \\ p_{m-1} \end{bmatrix} = - \begin{bmatrix} y_m \\ y_{m+1} \\ \vdots \\ y_{2m-1} \end{bmatrix}.$$

The coefficient matrix is $V_m \operatorname{diag}(c_r) V_m^T$, hence is nonsingular because the θ_r are distinct and the c_r are nonzero. Thus the polynomial coefficients are determined. Once p is known, its roots give the modes. The first m equations

$$y_n = \sum_{r=1}^{m} c_r \theta_r^n, \qquad 0 \le n < m,$$

form a Vandermonde system for the amplitudes. □

This theorem is classical Prony theory in the notation of takeoff kernels. In practice one uses a numerically stable variant: matrix pencil, ESPRIT, Hankel-SVD, or prediction-error fitting.

Corollary 1 (Minimal recurrence)**.** *Under the hypotheses of the theorem, y_n satisfies a unique minimal linear recurrence of order m,*

$$y_{n+m} = -\sum_{\ell=0}^{m-1} p_\ell y_{n+\ell},$$

and no recurrence of smaller order generates the same infinite sequence.

Proof. The recurrence follows from the annihilating polynomial. If a smaller recurrence existed, its characteristic polynomial would annihilate every θ_r, but a nonzero polynomial of degree less than m cannot have the m distinct roots $\theta_1, \dots, \theta_m$. □

Proposition 1 (Tail dominance and modal visibility)**.** *Let*

$$y_n = \sum_{r=1}^{m} c_r \theta_r^n,$$

where the modes θ_r are distinct and the amplitudes c_r are nonzero. Suppose the modes are ordered so that

$$|\theta_1| > \lambda, \qquad \lambda = \max_{r \ge 2} |\theta_r|.$$

Then the largest-radius mode is asymptotically visible:

$$y_n = c_1 \theta_1^n \left(1 + O\left(\left(\frac{\lambda}{|\theta_1|}\right)^n\right)\right).$$

More generally, for two modes r and s with $|\theta_r| > |\theta_s|$, mode r dominates mode s by a factor $\eta > 1$ once

$$n \ge \frac{\log(\eta |c_s| / |c_r|)}{\log(|\theta_r| / |\theta_s|)}.$$

When equality is used with $\eta = 1$, this gives the formal crossover index

$$n_{r,s} = \frac{\log(|c_s|/|c_r|)}{\log(|\theta_r|/|\theta_s|)}.$$

Proof. Factor out the largest-radius term:

$$y_n = c_1\theta_1^n \left(1 + \sum_{r=2}^{m} \frac{c_r}{c_1}\left(\frac{\theta_r}{\theta_1}\right)^n\right).$$

The parenthesized correction is

$$1 + O\left(\left(\frac{\lambda}{|\theta_1|}\right)^n\right),$$

because $|\theta_r/\theta_1| \le \lambda/|\theta_1| < 1$. For the pairwise visibility estimate, solve

$$|c_r|\,|\theta_r|^n \ge \eta|c_s|\,|\theta_s|^n$$

for n. This gives the displayed threshold. □

Remark 1. *The proposition is an asymptotic and pairwise visibility statement. In a finite window, a mode can still be hidden by boundary effects, noise, nearby modes, or cancellation among several comparable components. This is why the numerical procedure should report Hankel singular values, window sensitivity, and holdout prediction error rather than only a list of fitted modes.*

Kernel visibility. The same mode contributes to the takeoff kernel by finite difference:

$$\kappa_n = F_n - F_{n-1} \quad \rightsquigarrow \quad c_r(\theta_r - 1)\theta_r^{n-1}.$$

Thus very slow modes with $\theta_r \approx 1$ may be subtle in the kernel while still controlling the long settling of the profile F_n.

Oscillatory modes. If a conjugate pair has

$$\theta, \overline{\theta} = re^{\pm i\omega},$$

then its real contribution can be written

$$Cr^n \cos(\omega n + \phi).$$

Thus the fitted parameters have direct qualitative meaning: r is the settling rate, ω is the ringing frequency, and ϕ is the phase set by the boundary convention and early response. This is why the same modal language covers monotone smoothing, alternating overshoot, and near-periodic takeoff.

From modes to recurrence roots. For a finite-lag recurrence with denominator

$$Q(z) = 1 - \sum_{j=1}^{L} a_j z^j,$$

Planned Figure 3: modal visibility. Plot two or three components $|c_r||\theta_r|^n$ and their sum. Mark the crossover index $n_{r,s}$ where a weaker but slower-decaying component overtakes a stronger startup component. A second panel should show the same effect in the differenced kernel κ_n, where modes near $\theta = 1$ are attenuated by $\theta - 1$.

Figure 3: When response modes become visible in a composite takeoff curve.

let $\rho = e^{-\alpha}$ be the dominant positive root. A non-dominant singularity ζ_r contributes a mode

$$\theta_r = \frac{\rho}{\zeta_r}.$$

Thus modal recovery gives candidate singularities

$$\widehat{\zeta}_r = \frac{\widehat{\rho}}{\widehat{\theta}_r}.$$

If enough roots are recovered and the bounded-lag model is correct, these roots constrain Q, and therefore the combined coefficients a_j.

One reconstruction route is root-based. Fix a maximum lag L and write

$$Q_L(z) = 1 - \sum_{j=1}^{L} a_j z^j.$$

The ideal constraints are

$$Q_L(\rho) = 0, \qquad Q_L(\zeta_r) = 0 \quad \text{for recovered singularities } \zeta_r.$$

With noisy modes, replace them by a weighted residual:

$$\min_{a \in \mathbb{R}^L} \left|1 - \sum_{j=1}^{L} a_j \widehat{\rho}^j\right|^2 + \sum_r \omega_r \left|1 - \sum_{j=1}^{L} a_j \widehat{\zeta}_r^j\right|^2.$$

The weights ω_r should be smaller for unstable or weakly recovered modes. The solution is only a candidate denominator; it must still be checked against the original sequence and against structural constraints such as $a_j \ge 0$ or $a_j \in \mathbb{Z}_{\ge 0}$.

A second route avoids root reconstruction and fits the recurrence directly:

$$a^{(L)} = \arg\min_{a_1,\ldots,a_L} \sum_{n=L}^{T} \left(N_n - 1 - \sum_{j=1}^{L} a_j N_{n-j}\right)^2.$$

This direct fit is usually more stable numerically. The modal model then acts as an explanation and diagnostic for the fitted recurrence: its roots should produce the same observed damping, ringing, and crossover behavior.

Repeated roots and polynomial factors. If Q has repeated roots, the modal expansion contains terms

$$n^\ell \theta^n.$$

The first version of the follow-up should state the simple-root theory first, then add repeated roots as a standard extension.

Planned Figure 4: nonidentifiability. Show two different channel diagrams with the same combined coefficients $a_1 = 2, a_4 = 2$. Below them, plot the identical F_n and κ_n. The figure should make clear that response equivalence is exact, not a numerical accident.

Figure 4: Different source stories can induce the same takeoff response.

5 Nonidentifiability of Sources

Proposition 2 (Branch-channel decomposition is not identifiable)**.** *The latent split $a_j = b_j + c_j$ cannot be recovered from the tree sequence (N_n) without additional assumptions. Any two splits with the same combined coefficients a_j generate the same recurrence, the same tree sequence, and the same takeoff kernel.*

Proof. The recurrence depends on b_j and c_j only through their sum:

$$N_n = 1 + \sum_j (b_j + c_j) N_{n-j} = 1 + \sum_j a_j N_{n-j}.$$

Therefore all splits with the same a_j are observationally equivalent from the perspective of N_n, u_n, F_n, and κ_n. □

Equivalently, a single trace identifies at most an equivalence class

$$[(b, c)] = \{(b', c') : b'_j + c'_j = a_j \text{ for all } j\}.$$

Every element of this class has the same response. The labels "fast" and "slow" become meaningful only when they are tied to information not present in the single aggregate trace: separated lag supports, known code paths, interventions that disable one channel, or multiple measurements in which the channels are perturbed differently.

A simple example is

$$a_1 = 2, \qquad a_4 = 2.$$

The split

$$b_1 = 2,\ b_4 = 0, \qquad c_1 = 0,\ c_4 = 2$$

and the split

$$b_1 = 1,\ b_4 = 1, \qquad c_1 = 1,\ c_4 = 1$$

produce the same combined recurrence. One interpretation is a fast channel plus a slow channel; the other is two identical mixed channels. A single composite takeoff trace cannot choose between them.

Remark 2. *This is not a defect of the framework. It is the useful warning. Takeoff identifies response modes before it identifies source mechanisms.*

6 Constrained Structural Recovery

To make source reconstruction meaningful, impose structure. Natural constraints include:

1. bounded lag: $a_j = 0$ for $j > L$;
2. nonnegative branch counts: $a_j \geq 0$;
3. integer branch counts: $a_j \in \mathbb{Z}_{\geq 0}$;
4. sparsity: only a few lags are active;
5. separated supports for fast and slow channels;
6. multiple traces from controlled perturbations of the same program.

For a fixed maximum lag L, let $\mathcal{A}_L$ be the feasible class. For example,

$$\mathcal{A}_L^+ = \{a \in \mathbb{R}^L : a_j \geq 0\}, \qquad \mathcal{A}_L^{\mathbb{Z}} = \{a \in \mathbb{Z}_{\geq 0}^L\}.$$

A direct combined-recurrence fit is then

$$\widehat{a}^{(L)} = \arg\min_{a \in \mathcal{A}_L} \sum_{n=L}^{T} \left(N_n - 1 - \sum_{j=1}^{L} a_j N_{n-j} \right)^2 + \lambda \|a\|_1,$$

where $\lambda \|a\|_1$ encourages sparse active lags. A model-selection score can then penalize lag length and support size:

$$\begin{aligned} \mathrm{Score}(L, a) = \log \left(\frac{1}{T - L + 1} \sum_{n=L}^{T} R_n(a)^2 \right) \\ + \tau |\operatorname{supp}(a)| + \gamma L, \end{aligned}$$

where

$$R_n(a) = N_n - 1 - \sum_{j=1}^{L} a_j N_{n-j}.$$

For branch-count reconstruction, replace the nonnegative constraint by

$$a_j \in \mathbb{Z}_{\geq 0}$$

and solve a small mixed-integer least-squares problem or use nonnegative least squares followed by integer polishing.

If only the normalized profile is available, reconstruct a scale-free tree proxy by

$$\log \widetilde{N}_{n+1} = \log \widetilde{N}_n + \widehat{\alpha} \widehat{F}_n.$$

The multiplicative scale of $\widetilde{N}_n$ is arbitrary, but the velocity profile is preserved. This is weaker than observing N_n directly and should be treated as a diagnostic rather than a definitive reconstruction.

Forward validation. Every candidate $\widehat{a}$ should be pushed back through the forward map. Generate $\widehat{N}_n$ from

$$\widehat{N}_n = 1 + \sum_{j=1}^{L} \widehat{a}_j \widehat{N}_{n-j}$$

using the same boundary convention as the data, and compare the induced $\widehat{F}_n$, $\widehat{\kappa}_n$, and modal roots against the observed ones. A recurrence that fits N_n but produces the wrong takeoff shape is a poor explanation, even if its one-step residual is small.

Channel recovery with separated supports. Assume two channels

$$a = b + c,$$

with

$$\begin{aligned} \mathrm{supp}(b) &\subseteq \{1, \dots, L_f\}, \\ \mathrm{supp}(c) &\subseteq \{L_s, \dots, L\}, \\ L_f &< L_s. \end{aligned}$$

Then a fast/slow interpretation becomes more stable: early lags must be assigned to b, late lags to c, and only the gap region remains ambiguous. This is a plausible assumption for algorithms where one mechanism recurs locally and another returns after a large delayed dependency.

With multiple traces, the constraint can be stronger. Suppose experiments $e = 1, \dots, E$ change the channel weights but preserve supports:

$$a^{(e)} = s_e b + t_e c.$$

If the scalars s_e, t_e are known or separately estimable and the design matrix (s_e, t_e) has rank two, then the combined coefficients across experiments can separate b and c. This is the inverse-takeoff analogue of adding interventions or sensors: source recovery requires more than one aggregate view.

7 Algorithms

Algorithm 1: modal inverse takeoff.
1. Measure N_n and, if relevant, M_n.
2. Estimate α from a stable tail window of u_n.
3. Form $Y_n = \widehat{F}_n - 1$, optionally after discarding a warmup prefix.
4. Choose model order m from Hankel singular values, information criteria, or holdout prediction error.
5. Estimate modes using Prony, matrix pencil, ESPRIT, or Hankel-SVD.
6. Convert complex conjugate pairs to damping, frequency, and phase:

$$\theta = re^{i\omega} \quad \rightsquigarrow \quad Cr^n \cos(\omega n + \phi).$$

7. Validate by forward-predicting F_n, κ_n, and the settling envelope.

The modal report should include the table

$$\{(|\theta_r|, \arg \theta_r, c_r, n_{r,s})\},$$

where crossover entries $n_{r,s}$ are reported only for pairs whose amplitudes are large enough to be visible in the observation window. This prevents the method from over-reading tiny fitted poles.

Algorithm 2: constrained recurrence search.
1. Pick a maximum lag $L_{\max}$.
2. For each $L \le L_{\max}$, fit $a_1, \dots, a_L$ under nonnegative or integer constraints.
3. Simulate the recovered recurrence and compare its takeoff profile to the observed profile.
4. Penalize large L, dense supports, negative implied modes, and unstable dominant roots.
5. Report a small set of candidate recurrences rather than a single overconfident source mechanism.

Algorithm 3: channel interpretation.
1. Start from a recovered combined recurrence $\widehat{a}$.
2. Declare the external assumptions: separated supports, known code paths, intervention labels, or multiple traces.
3. Solve for channel coefficients only inside the assumed class.
4. Report the equivalence class left unresolved by the assumptions.

This third step is intentionally conservative. It is where explanatory stories enter, so it should be the most explicit about assumptions.

8 Examples and Experiments

The first experiments should be deliberately simple. Their purpose is not to beat existing algorithms for recurrence solving. Their purpose is to test whether inverse takeoff recovers the right qualitative structure from finite traces: immediate response, delay, monotone settling, oscillation, mixed-stage response, and failure on non-exponential tails.

Each synthetic experiment should report:

$$\widehat{m}, \qquad \{(\widehat{\theta}_r, \widehat{c}_r)\}_{r=1}^{\widehat{m}}, \qquad \widehat{a},$$

along with the residuals

$$\sum_n |Y_n - \widehat{Y}_n|^2, \qquad \sum_n |F_n - \widehat{F}_n|^2, \qquad \sum_n |\kappa_n - \widehat{\kappa}_n|^2.$$

The first residual tests modal fit; the second and third test whether the shape of redundancy onset is reproduced.

Fibonacci. The recurrence

$$N_n = 1 + N_{n-1} + N_{n-2}$$

has a monotone leading correction under node counting and a more visible alternating component under leaf counting. This tests sensitivity to boundary and counting convention.

Delayed binary recursion. The recurrence

$$N_n = 1 + 2N_{n-L}$$

has dead time in the original coordinate and immediate takeoff in the coarse coordinate $m = n/L$. This tests whether the inverse method recognizes delay.

Mixed fast/slow channels. Use

$$N_n = 1 + b_1 N_{n-1} + c_L N_{n-L}.$$

The intended response is a fast initial rise followed by a slow settling tail. This is the central synthetic case for the follow-up.

Planned Figure 5: synthetic benchmark grid. Rows: Fibonacci, delayed binary, mixed fast/slow, near-periodic slow family, and lattice-path negative control. Columns: F_n, κ_n, Hankel singular spectrum, recovered poles in the complex plane, and forward simulation of the best candidate recurrence.

Figure 5: Synthetic inverse-takeoff benchmark layout.

Planned Table 1: reconstruction metrics. For each benchmark, report true active lags, recovered active lags, modal order, pole error, holdout error, and whether the source decomposition is identified, underidentified, or deliberately not attempted.

Figure 6: Experimental analysis table reserved for the completed study.

Near-periodic slow family. Use the family

$$N_n = 1 + (2^L - 1)N_{n-L} + 2N_{n-L-1}.$$

The dominant speed is $\log 2$, but non-dominant roots approach the dominant circle and create a long ringing response. This tests modal resolution when roots are close to the unit circle in normalized coordinates.

Grid dynamic programs as a negative control. Diagonal lattice paths satisfy

$$P_{n,n} = \binom{2n}{n}$$

and have algebraic rather than exponential settling. A finite-exponential inverse model should either fail gracefully or require too many modes. This separates finite-lag recurrence takeoff from diffusive takeoff.

Real recursive traces. Instrument recursive programs to record:

$$N_n, \qquad M_n, \qquad u_n, \qquad w_n, \qquad v_n.$$

Candidate examples include memoized search, context-free parsing recurrences, edit-distance-style dynamic programs, and recursive enumeration tasks with changing branching factors.

For real traces, the paper should be careful about interpretation. The right claim is not that the recovered modes reveal the source code uniquely. The claim is that a measured recursive workload has a reproducible response shape. The analysis should therefore compare runs across input families, random seeds, and boundary conventions. A mode that persists across these choices is a stronger empirical object than a mode visible in only one fitted window.

9 Evaluation Criteria

The follow-up should avoid claiming that a recovered modal model is the true program mechanism. Report several levels of evidence:

1. modal residual:
$$\sum_n |Y_n - \widehat{Y}_n|^2;$$
2. Hankel-rank stability under window shifts;
3. pole stability under bootstrap or subsampling;
4. recurrence residual on N_n;
5. holdout prediction error for later n;
6. stability of inferred support $\mathrm{supp}(a)$;
7. sensitivity to warmup cutoff and base-case convention.

Failure modes should be part of the story: short windows, nearly colliding modes, noisy counts, algebraic tails, hidden DAG cancellation, and multiple structural decompositions with the same combined coefficients.

A useful reporting convention is to classify each run at three levels:

$$\begin{aligned} &\text{modal fit} \Rightarrow \text{combined recurrence} \\ &\Rightarrow \text{source interpretation.} \end{aligned}$$

The modal fit is accepted when it predicts held-out Y_n and has stable Hankel rank. The combined recurrence is accepted when it reproduces both N_n and the takeoff shape. The source interpretation is accepted only when the stated assumptions make the decomposition identifiable. This three-level reporting keeps a successful signal fit from being mistaken for a recovered program mechanism.

10 Related Literature Map

Analytic combinatorics and recurrence asymptotics. The forward theory is the standard rational-generating-function story: singularities of the generating function determine exponential growth and subdominant correction modes. This is the basis for translating finite-lag recurrences into takeoff modes.

Prony, matrix pencil, and ESPRIT. These methods recover sums of exponentials and damped sinusoids from samples. They are the closest technical literature for the exact and noisy modal inverse takeoff problem.

Berlekamp-Massey, Pade approximation, and minimal recurrences. For exact linearly recurrent sequences, one can recover a shortest recurrence or rational generating function from finite samples. This is the algebraic version of the inverse recurrence problem.

System identification. Prediction-error methods, subspace identification, Ho-Kalman realization, and Hankel-SVD methods provide the engineering language: infer a small dynamical system from an observed impulse or step response.

Blind source separation. This literature is relevant mostly as a caution. A single composite trace does not determine independent latent sources without additional assumptions, such as multiple sensors, independence, non-Gaussianity, interventions, or structural constraints.

Sparse inverse problems. Sparsity and regularization give a practical path from many possible recurrences to a small set of interpretable candidates.

11 Claim Map for the Complete Version

The complete paper should separate proved algebraic claims from empirical and algorithmic claims.

Exact algebraic claims. The finite-exponential model gives exact modal recovery from $2m$ samples, Hankel rank m, a minimal recurrence of order m, tail dominance of the largest-radius mode, and pairwise crossover estimates. These are the stable mathematical core of the paper.

Structural claims. Recovered modes determine candidate recurrence roots by

$$\widehat{\zeta}_r = \widehat{\rho}/\widehat{\theta}_r,$$

and bounded-lag recurrence candidates can be fit either from these roots or directly from N_n. The structural claims should be phrased as constrained reconstruction, not as unique source recovery.

Negative claims. Branch-channel splits $a = b + c$ are not identifiable from one aggregate trace. This result should remain explicit because it protects the inverse theory from overinterpretation.

Empirical claims. The experiments should show that synthetic recurrences are recovered in the right regime, that algebraic or diffusive tails trigger failure modes, and that real recursive traces have reproducible response shapes. Perturbation bounds for noisy modal recovery can be quoted from matrix-pencil or subspace-identification literature rather than reproved.

12 Conclusion

The inverse takeoff problem is best framed as a measured-response problem, not as a promise to recover hidden branches uniquely. In the finite-lag setting, the observed redundancy response can often be decomposed into modes:

$$F_n - 1 \approx \sum_r c_r \theta_r^n.$$

Those modes explain damping, ringing, delay, and settling. They also constrain possible recurrence mechanisms. The mode with largest radius becomes visible in the tail, and pairwise crossover estimates describe when slower-decaying modes overtake stronger short-lived components.

The subtle point is that this success stops one step short of source recovery. The same combined recurrence can support several incompatible stories about fast and slow channels. That ambiguity is not an embarrassment; it is the line between what the trace says and what an analyst adds. The trace says which response modes are present. Extra assumptions say which mechanisms may have produced them.

The remaining work is empirical as much as algebraic. The algebraic core gives Hankel rank, exact modal recovery, minimal recurrence, visibility thresholds, and nonidentifiability. The experimental core must show when these statements survive finite windows, noisy counts, boundary conventions, and real recursive programs. If that evidence is supplied, inverse takeoff kernels become a practical diagnostic: a way to read recursive redundancy as a response signal before trying to name its sources. The guiding principle is therefore:

Takeoff identifies response modes before it identifies source mechanisms.

References

[1] R. Bellman. *Dynamic Programming*. Princeton University Press, 1957.

[2] P. Flajolet and R. Sedgewick. *Analytic Combinatorics*. Cambridge University Press, 2009.

[3] Y. Hua and T. K. Sarkar. Matrix pencil method for estimating parameters of exponentially damped/undamped sinusoids in noise. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 38(5):814–824, 1990.

[4] L. Ljung. *System Identification: Theory for the User*. Prentice Hall, second edition, 1999.

[5] D. Potts and M. Tasche. Parameter estimation for nonincreasing exponential sums by Prony-like methods. *Linear Algebra and its Applications*, 439(4):1024–1039, 2013.

[6] G. de Prony. Essai experimental et analytique. *Journal de l'Ecole Polytechnique*, 1(2):24–76, 1795.

[7] P. Van Overschee and B. De Moor. *Subspace Identification for Linear Systems*. Kluwer Academic Publishers, 1996.

[8] A. Hyvarinen and E. Oja. Independent component analysis: algorithms and applications. *Neural Networks*, 13(4–5):411–430, 2000.